

Generative AI (AI4009)

Sessional-II Exam

Date: April 14th 2025

Course Instructor

Dr. Akhtar Jamil

Total Time (Hrs): 1

Total Marks: 50

Total Questions: 3

Roll No

Section

Student Signature

Do not write below this line

Attempt all the questions.

[CLO 1-3]

Question No. 1. MCQ [1 x 25 = 25]

Answer the MCQs on the given answer sheet attached at the end of the question paper. Answers marked on the question paper will not be evaluated.

1. Why does the KV cache become a bottleneck in long-context inference?
A. It increases FLOPs quadratically
B. It requires recomputation of past tokens
C. It causes expensive data movement between HBM and SRAM
D. It limits attention head parallelism
2. If Top-k routing is changed from k=2 to k=8, what will be the main effect of this change?
A. Lower training stability
B. Higher communication and aggregation overhead
C. Reduced model capacity
D. Elimination of routing collapse
3. In DeepSeekMoE, what is the key functional difference between shared experts and routed experts?
A. Shared experts are updated less frequently
B. Routed experts process all tokens uniformly
C. Shared experts capture general patterns, routed experts specialize
D. Routed experts are used only during inference
4. If AdaIN is removed from StyleGAN but everything else remains the same, what is the most likely effect?
A. Improved stochastic variation
B. Loss of layer-wise control over styles
C. Reduced memory usage
D. Faster convergence

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5. In the Transformer architecture, the scaled dot-product attention uses a scaling factor of $1/\sqrt{d_k}$. If this factor were removed while keeping a high dimensionality d_k , what is the most likely logical failure mode for the model during the initial phases of training?
- A) **It may lead to vanishing gradients.**
 - B) The attention mechanism may not be utilizing positional encodings effectively.
 - C) The Queries and Keys would become linearly dependent, forcing the attention matrix to become a constant identity matrix.
 - D) The model would behave normal as usual.
6. In StyleGAN, what would happen if noise inputs were removed from all layers?
- A. Global structure will collapse
 - B. Training becomes unstable
 - C. Latent space becomes more entangled
 - D. **Images will lose fine-grained stochastic variations**
7. Which limitation of RNNs directly motivated the design of Transformers?
- A. Lack of embeddings
 - B. Inability to model sequences
 - C. **Sequential computation preventing parallelization**
 - D. High memory efficiency
8. If we remove the encoder and keep only the decoder in a Transformer, what functionality is lost?
- A. Ability to generate sequences
 - B. **Ability to attend to input sequence representations**
 - C. Ability to compute attention weights
 - D. Ability to use embeddings
9. Why is masking applied in the Transformer decoder's self-attention?
- A. To reduce computation
 - B. To prevent attending to padding tokens only
 - C. To stabilize gradients
 - D. **To prevent the model from seeing future tokens during training**
10. StyleGAN start from a learned constant input instead of directly using the latent vector as input for image generation.
- A. **True**
 - B. False
11. Let $q = [2, 0]$, $k_1 = [2, 0]$, $k_2 = [0, 2]$.
Using scaled dot-product attention with $d_k = 2$, what is the scaled score for k_1 without softmax?
- A. 4
 - B. $2\sqrt{2}$
 - C. 2
 - D. $4/\sqrt{2}$

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12. In StyleGAN, why does mapping the latent vector from Z to W improve interpolation between image features?
- A. It increases the size of the latent vector
 - B. It removes the need for noise injection
 - C. It allows direct input to the synthesis network
 - D. It creates a more linear and disentangled feature space**
13. In BERT, why are segment embeddings used even though a [SEP] token separates Sentence A and Sentence B?
- A. To help positional encoding track distances
 - B. To enable next token prediction
 - C. To tell each token which sentence it belongs to**
 - D. To hide tokens during masked language modeling
14. Suppose a Vision Transformer divides an image into 200 equal size patches. What will be the total input sequence length to the encoder?
- A. 199
 - B. 200
 - C. 201**
 - D. 202
15. In DeepSeekMoE, the bias update rate (γ) influences expert selection probabilities. What happens when γ is set too high?
- A. All experts are selected equally
 - B. Selection probabilities change too rapidly between steps**
 - C. Experts stop receiving tokens
 - D. Routing becomes deterministic
16. In StyleGAN, AdaIN applies scaling (y_s) and bias (y_b). If the scaling term y_s is removed, what is the most likely effect?
- A. Spatial structure remains, but feature intensity control (e.g., contrast, color strength) is lost**
 - B. Image collapses to a uniform color defined by y_b
 - C. The synthesis network stops at low resolution
 - D. Noise becomes deterministic
17. A sequence has 4 tokens in the decoder with causal masking. How many tokens can token 3 attend to?
- A. 1
 - B. 2
 - C. 3**
 - D. 4

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18. A Transformer encoder block uses a feed-forward network (FFN) with two linear layers configured as follows:

- Input dimension: 32
- Hidden dimension: 128
- Output dimension: 32

Each linear layer includes one bias term per output neuron. How many total **learnable parameters** are in this FFN?

- A) 8,192
- B) 8,256
- C) 8,352**
- D) 8,384

19. In DeepSeek, what does context extension mean?

- A) Activating more experts per token
- B) Allowing longer input sequences**
- C) Reducing training parameters
- D) Using RLHF for alignment

20. In a Mixture-of-Experts (MoE) layer, the Router plays a crucial role. Which of the following statements best describes its function?

- A) It normalizes the hidden vector using RMSNorm before passing it to the experts
- B) It reduces the dimensionality of the input vector to minimize computation
- C) It computes scores for token and removes the top-K tokens from the input sequence
- D) It computes scores for experts and selects the top-K experts to process each token**

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Question No 2.

Write short answers to the following questions. [3 x 5=15]

Q.1. In Mixture-of-Experts (MoE) models, what is routing collapse, and how does the architecture of DeepSeek-V3 mitigate this issue during training?

Routing collapse happens when, in a Mixture-of-Experts model, most of the input is sent to only a few experts while others are hardly used, which wastes the model's potential. In DeepSeek-V3, this issue is mitigated through a combination of techniques including load balancing loss to encourage uniform expert usage, top-k routing (such as assigning each token to multiple experts) to improve diversity, auxiliary regularization to discourage over-reliance on specific experts, and capacity constraints to limit the number of tokens processed by each expert.

Q.2. What is mixing regularization in StyleGAN, and what advantage does it provide?

Mixing regularization uses multiple latent codes at different layers of the generator instead of a single latent vector. It improves disentanglement and increases diversity in generated images by encouraging the model to learn independent styles at different layers and avoid reliance on a single latent code.

Q.3. Why is LLM inference often memory-bound rather than compute-bound, and how does the KV cache contribute to this?

LLM inference is memory-bound because data movement is slower than computation. The KV cache grows with each generated token and must be repeatedly transferred from VRAM (HBM) to fast on-chip memory (SRAM) for attention computation. Since this data transfer is slower than the compute speed of tensor cores, it becomes the main bottleneck.

Q.4. What is the role of the [CLS] token in BERT?

The [CLS] token acts as a global representation of the entire input sequence. Its final embedding is used for downstream tasks such as classification.

Q.5. Why is Multi-Token Prediction (MTP) used during training in DeepSeek-V3 even though it is not used during inference?

MTP is used to improve the quality of learned representations. It encourages the model to capture future dependencies and encode richer contextual information in the current hidden state, acting as a form of regularization.

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Question No 3. Long questions. [5 + 10 =15]

Q.1. **Suppose** $x = [3, 3, 3]$ and $\gamma = [0.5, 0.5, 0.5]$

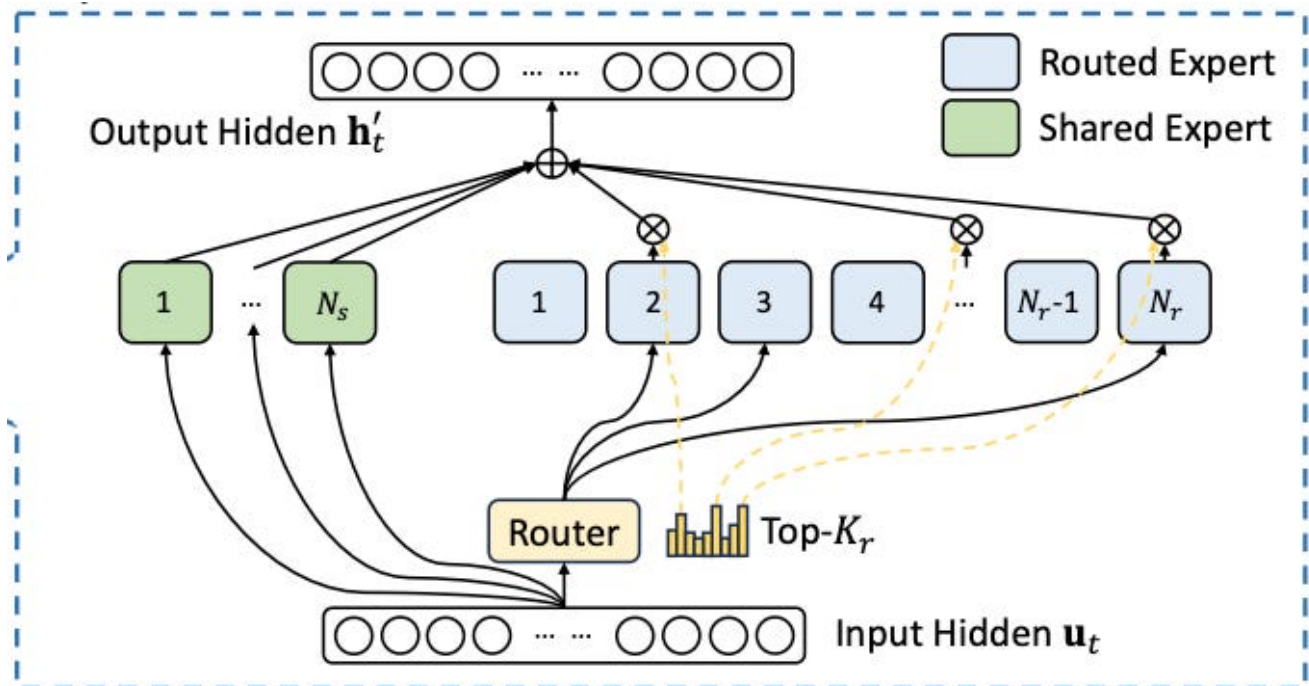
Compute the RMSNorm output vector as used in DeepSeek v3 paper.

- Step 1: Compute mean of squares: $(3^2 + 3^2 + 3^2)/3 = (9 + 9 + 9)/3 = 27/3 = 9$
- Step 2: $\sqrt{(\text{mean}(x^2))} = \sqrt{9} = 3$
- Step 3: Normalize: $[3/3, 3/3, 3/3] = [1, 1, 1]$
- Step 4: RMSNorm: $[1, 1, 1] \times [0.5, 0.5, 0.5] = [0.5, 0.5, 0.5]$

Q.2. With the help of a diagram, explain the working of the Mixture of Experts (MoE) mechanism in the DeepSeek-V3 model.

In DeepSeek-V3, the Mixture of Experts (MoE) replaces the standard feed-forward layer with multiple expert networks. For each input token, a router computes scores for all experts and selects the top-k experts based on these scores. Only the selected experts process the token, and their outputs are combined using weighted aggregation according to the router's probabilities.

This approach allows the model to activate only a small subset of experts per token, increasing model capacity without significantly increasing computation. DeepSeek-V3 also includes shared experts for general knowledge and routed experts for specialized tasks, while load balancing ensures that all experts are utilized effectively during training.



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Q.3. Given the sentence “**Deep learning models work**”. Its embedding representation is:

$$X = \begin{bmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 1 & 0 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 & 1 & 2 \\ 4 & 3 & 2 & 1 & 0 & 1 \end{bmatrix}$$

Apply Rotary Positional Encoding (RoPE) to the word at **position = 1** (“**learning**”) using $d = 6$ to compute the position vector.

The word at position 1 is “learning”, so its vector is

$$x = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

We split it into three pairs:

$$x^{(0)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad x^{(1)} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad x^{(2)} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}$$

For RoPE,

$$\theta_i = \frac{p}{10000^{\frac{2i}{d}}}$$

with $p = 1$ and $d = 6$.

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$$\theta_0 = \frac{1}{10000^{\frac{0}{6}}} = 1$$

$$\theta_1 = \frac{1}{10000^{\frac{2}{6}}} = \frac{1}{10000^{1/3}} \approx 0.0464$$

$$\theta_2 = \frac{1}{10000^{\frac{4}{6}}} = \frac{1}{10000^{2/3}} \approx 0.00215$$

Now compute the rotation matrices

$$R(\theta_i) = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{bmatrix}$$

For $i = 0$,

$$R(\theta_0) = \begin{bmatrix} \cos(1) & -\sin(1) \\ \sin(1) & \cos(1) \end{bmatrix} = \begin{bmatrix} 0.5403 & -0.8415 \\ 0.8415 & 0.5403 \end{bmatrix}$$

For $i = 1$,

$$R(\theta_1) = \begin{bmatrix} \cos(0.0464) & -\sin(0.0464) \\ \sin(0.0464) & \cos(0.0464) \end{bmatrix} \approx \begin{bmatrix} 0.9989 & -0.0464 \\ 0.0464 & 0.9989 \end{bmatrix}$$

For $i = 2$,

$$R(\theta_2) = \begin{bmatrix} \cos(0.00215) & -\sin(0.00215) \\ \sin(0.00215) & \cos(0.00215) \end{bmatrix} \approx \begin{bmatrix} 0.999998 & -0.00215 \\ 0.00215 & 0.999998 \end{bmatrix}$$

Now multiply each rotation matrix with its pair.

For the first pair,

$$R(\theta_0)x^{(0)} = \begin{bmatrix} 0.5403 & -0.8415 \\ 0.8415 & 0.5403 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 2(0.5403) - 1(0.8415) \\ 2(0.8415) + 1(0.5403) \end{bmatrix} = \begin{bmatrix} 0.2391 \\ 2.2233 \end{bmatrix}$$

For the second pair,

$$R(\theta_1)x^{(1)} = \begin{bmatrix} 0.9989 & -0.0464 \\ 0.0464 & 0.9989 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -0.0464 \\ 0.9989 \end{bmatrix}$$

For the third pair,

$$R(\theta_2)x^{(2)} = \begin{bmatrix} 0.999998 & -0.00215 \\ 0.00215 & 0.999998 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2(0.999998) - 3(0.00215) \\ 2(0.00215) + 3(0.999998) \end{bmatrix} = \begin{bmatrix} 1.9935 \\ 3.0043 \end{bmatrix}$$

Therefore, the final RoPE representation of "learning" is

$$\begin{bmatrix} 0.2391 \\ 2.2233 \\ -0.0464 \\ 0.9989 \\ 1.9935 \\ 3.0043 \end{bmatrix}$$

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Answer Sheet MCQs

Fill the correct option. Only one option must be selected. Selection of multiple options or overwriting will result in ZERO marks.

CORRECT METHOD		WRONG METHOD	
☐	☒	☒	☒
NAME:		ROLL NO:	
Roll No	A B C D		
0	☐	☐	☐
1	☐	☐	☐
2	☐	☐	☐
3	☐	☐	☐
4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐
MCQs		A B C D	
Section1		A B C D	
1	☐	☐	☐
2	☐	☐	☐
3	☐	☐	☐
4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐
10	☐	☐	☐